

Month	Observation	Deviation from the Mean	Squared Deviation
September	3	0.75	0.5625
October	0	-2.25	5.0625
November	6	3.75	14.0625
December	5	2.75	7.5625
Sum	27		96.2500

The sample standard deviation is $\sqrt{\frac{96.2500}{11}} = 2.958\%$.

2. Given the data in Example 3, calculate the target downside deviation if the target is 2 percent.

Solution:

Month	Observation	Deviation from the 2% Target	Deviations below the Target	Squared Deviations below the Target
January	5	3	—	—
February	3	1	—	—
March	-1	-3	-3	9
April	-4	-6	-6	36
May	4	2	—	—
June	2	0	—	—
July	0	-2	-2	4
August	4	2	—	—
September	3	1	—	—
October	0	-2	-2	4
November	6	4	—	—
December	5	3	—	—
Sum				53

The target semideviation with 2 percent target = $\sqrt{\frac{53}{11}} = 2.195$ percent.

3. Compare the standard deviation, the target downside deviation if the target is 2 percent, and the target downside deviation if the target is 3 percent.

Solution:

The standard deviation is based on the deviation from the mean, or 27% / 12 = 2.25%. The standard deviation includes all deviations from the mean, not just those below it. This results in a sample standard deviation of 2.958 percent.

Considering just the four observations below the 2 percent target, the target semideviation is 2.195 percent. It is less than the sample standard deviation since target semideviation captures only the downside risk (i.e., deviations below the target). Considering target semideviation with a 3 percent target, there are now five observations below 3 percent, so the target semideviation is higher, at 2.763 percent.